Complex Analysis Homework 6

1. (a) Suppose that $f(z)$ is analytic on and inside the closed disk $|z| \le R$ and that $f(0) = 0$.

(i) Prove that the function

$$F(z) := \begin{cases} f(z)/z & z \neq 0 \\ f'(0) & z = 0 \end{cases}$$

is analytic on $|z| \le R$.

(ii) Suppose $|f(z)| \le M = \max_{|\omega|=R}\{|f(\omega)|\}$ inside the circle $|z| = R$. Show that

$$|f(z)| \le \frac{M}{R}|z| \quad \text{and} \quad |f'(0)| \le \frac{M}{R},$$

and equality holds only if $f(z) = \frac{M}{R}e^{i\theta}z$, where θ is real.

(b) Suppose $f(z)$ is analytic on and inside the unit circle $|z| = 1$, and that $\text{Re}\, f(z) > 0$, $f(0) = a > 0$, show that $\left|\frac{f(z)-a}{f(z)+a}\right| \le |z|$ and $|f'(0)| \le 2a$.

Hint: Consider the function $g(z) = \frac{z-a}{z+a}$. Show that $|g(z)| < 1$ if $\text{Re}(z) > 0$.

Solution

(a)(i) By Theorem 15 in P.207, $G(z) = \frac{1}{2\pi i}\int_{|\zeta|=R} \frac{f(\zeta)}{\zeta(\zeta - z)}d\zeta$ is analytic for all z not on $|z| = R$.

$$G(0) = \frac{1}{2\pi i}\int_{|\zeta|=R} \frac{f(\zeta)}{\zeta^2}d\zeta = f'(0) = F(0)$$

For $z \neq 0$, $G(z) = \frac{1}{2\pi i}\int_{|\zeta|=R} \frac{f(\zeta)}{\zeta(\zeta - z)}d\zeta$

$$= \frac{1}{z}\left[\frac{1}{2\pi i}\int_{|\zeta|=R} \frac{f(\zeta)}{\zeta - z}d\zeta - \frac{1}{2\pi i}\int_{|\zeta|=R} \frac{f(\zeta)}{\zeta}d\zeta\right]$$

$$= \frac{1}{z}(f(z) - f(0))$$

$$= F(z)$$

Therefore $G(z) = F(z)$ for any z in $|z| < R$, and $F(z)$ is analytic on $|z| \le R$.

(a)(ii) $|f(z)| \le \frac{M}{R}|z|$ if $z = 0$.

$$|F(z)| \le \max_{|\omega|=R}\{|F(\omega)|\} = \max_{|\omega|=R}\left\{\left|\frac{f(\omega)}{\omega}\right|\right\} = \frac{M}{R}$$

$$\left|\frac{f(z)}{z}\right| \le \frac{M}{R} \quad \text{for } z \ne 0, \quad \text{and} \quad |f'(0)| \le \frac{M}{R}$$

Hence $|f(z)| \le \frac{M}{R}|z|$ and $|f'(0)| \le \frac{M}{R}$.

Let α be a complex number such that $|\alpha| = R$ and $|f(\alpha)| = \max_{|\omega|=R}\{|f(\omega)|\} = M$

Equality holds only if $F(z)$ is constant. Then $\left|\frac{f(z)}{z}\right| = |F(z)| = |F(\alpha)| = \left|\frac{f(\alpha)}{\alpha}\right| = \frac{M}{R}$ for $z \ne 0$

Hence $f(z) = \frac{M}{R}e^{i\theta}z$.

(b) $|z-a|^2 = (x-a)^2 + y^2 = x^2 - 2ax + a^2 + y^2 \overset{a,x>0}{\le} x^2 + 2ax + a^2 + y^2 = |z+a|^2$

$$|g(z)| = \left|\frac{z-a}{z+a}\right| < 1.$$

$g(f(z))$ is analytic on and inside the closed disk $|z| \le 1 = R$. $|g(f(z))| < 1 = M$ and $g(f(0)) = 0$. By (a)(ii),

$$\left|\frac{f(z)-a}{f(z)+a}\right| = |g(f(z))| \le \frac{M}{R}|z| = |z| \quad \text{and} \quad |(g \circ f)'(0)| \le \frac{M}{R}.$$

$$(g \circ f)'(0) = g'(f(0))\,f'(0) = g'(a)\,f'(0)$$

$$g'(a) = \left.\frac{(z+a)-(z-a)}{(z+a)^2}\right|_{z=a} = \frac{2a}{(2a)^2} = \frac{1}{2a}$$

$$|(g \circ f)'(0)| \le \frac{M}{R} \quad \Rightarrow \quad \frac{1}{2a}|f'(0)| \le 1 \quad \Rightarrow \quad |f'(0)| \le 2a$$

2. Suppose f is analytic in $|z| \le 1$. Suppose $|f(z)| \le 2$ for $z = e^{i\theta},\ 0 < \theta \le \pi$ and $|f(z)| \le 3$ for $z = e^{i\theta},\ -\pi < \theta \le 0$. Show that $|f(0)| \le \sqrt{6}$. [Hint: Consider $f(z) \cdot f(-z)$.]

Solution

Let $g(z) = f(z) \cdot f(-z)$. g is analytic in $|z| \le 1$.

If $z = e^{i\theta},\ 0 < \theta \le \pi$ then $|f(z)| \le 2$ and $|f(-z)| \le 3$.

If $z = e^{i\theta},\ -\pi < \theta \le 0$ then $|f(z)| \le 3$ and $|f(-z)| \le 2$.

In conclusion, $|g(z)| = |f(z) \cdot f(-z)| \le 6$ if $|z| = 1$

$$g(0) = \frac{1}{2\pi i}\int_{|z|=1} \frac{g(z)}{z}\,dz$$

$$|f(0) \cdot f(-0)| = |g(0)| \le \frac{1}{2\pi}\frac{6}{1}2\pi = 6$$

$$|f(0)| \le \sqrt{6}$$

3. (a) Let α be a complex number. Show that if $(1+z)^{\alpha}$ is taken as $e^{\alpha \mathrm{Log}(1+z)}$, then for $|z| < 1$

$$(1+z)^{\alpha} = 1 + \frac{\alpha}{1}z + \frac{\alpha(\alpha-1)}{2\cdot 1}z^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3\cdot 2\cdot 1}z^3 + \cdots.$$

(b) Give the Taylar series expansion of $\sqrt{z}$ at $z_0 = 1$ and find its radius of convergence.

Solution

(a) $f(z) = (1+z)^\alpha$ is analytic on $|z| < 1$, with
$f(0) = 1$ and, for $j \geq 1$,
$f^{(j)}(z) = \alpha(\alpha-1)\cdots(\alpha-j+1)(1+z)^{\alpha-j}$
$f^{(j)}(0) = \alpha(\alpha-1)\cdots(\alpha-j+1)$
The Maclaurin series for $f(z)$ on $|z| < 1$ is

$$\sum_{j=0}^{\infty} \frac{f^{(j)}(0)}{j!} z^j = 1 + \sum_{j=1}^{\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-j+1)}{j!} z^j$$

(b) *From the definition of numbers with rational exponents $z^a = \exp(a \log z)$ we see that the square root cannot be analytic on all of $\mathbb{C}$. In fact, taking the first derivatives we see that $f^{(n)}(z) = 2^{-n} \prod_{k=0}^{n-1}(1-2k) z^{-n+\frac{1}{2}}$ and therefore*

$$a_n = \frac{1}{n!} f^{(n)}(1) = \frac{1}{n! 2^n} \prod_{k=0}^{n-1} (1-2k),$$

and with these a_n,

$$f(z) = \sqrt{z} = \sum_{n=0}^{\infty} a_n (z-1)^n = \sum_{k=0}^{\infty} \frac{1}{n! 2^n} \prod_{k=0}^{n-1} (1-2k)(z-1)^n.$$

Note that the logarithm is undefined at zero. Thus, of all branch cuts of the logarithm, the best radius of convergence we can achieve is 1.

4. Find the radius of convergence of the following power series:

(a) $\sum_{k=0}^{\infty} 2^k z^k$

(b) $\sum_{k=0}^{\infty} \frac{k}{6} z^k$

(c) $\sum_{k=1}^{\infty} k^2 z^k$

(d) $\sum_{k=0}^{\infty} \frac{3^k z^k}{4^k + 5^k}$

(e) $\sum_{k=1}^{\infty} \frac{2^k z^{2k}}{k^2 + k}$

(f) $\sum_{k=1}^{\infty} \frac{z^{2k}}{4^k k^k}$

(g) $\sum_{k=1}^{\infty} \frac{k^k}{1+2^k k^k} z^k$

Solution

(a) We apply the ratio test, thus

$$R = \lim_{k\to\infty} \left|\frac{a_k}{a_{k+1}}\right| = \lim_{k\to\infty} \left|\frac{2^k}{2^{k+1}}\right| = \lim_{k\to\infty} \frac{1}{2} = \frac{1}{2}.$$

(b) We apply the ratio test, thus

$$R = \lim_{k\to\infty} \left|\frac{a_k}{a_{k+1}}\right| = \lim_{k\to\infty} \left|\frac{k}{6}\frac{6}{k+1}\right| = \lim_{k\to\infty} \left|\frac{k}{k+1}\right| = \lim_{k\to\infty} \frac{1}{1+1/k} = 1.$$

(c) We apply the ratio test, thus

$$R = \lim_{k\to\infty} \left|\frac{a_k}{a_{k+1}}\right| = \lim_{k\to\infty} \left|\frac{k^2}{(k+1)^2}\right| = \lim_{k\to\infty} \frac{k^2}{k^2+2k+1} = \lim_{k\to\infty} \frac{1}{1+2/k+1/k^2} = 1.$$

(d) We apply the ratio test, thus

$$R = \lim_{k\to\infty} \left|\frac{a_k}{a_{k+1}}\right| = \lim_{k\to\infty} \left|\frac{3^k}{4^k+5^k}\frac{4^{k+1}+5^{k+1}}{3^{k+1}}\right| =$$

$$= \lim_{k\to\infty} \left|\frac{3^k \cdot 5^{k+1}\left(\left(\frac{4}{5}\right)^{k+1}+1\right)}{3^{k+1}\cdot 5^k\left(\left(\frac{4}{5}\right)^k+1\right)}\right| = \lim_{k\to\infty} \left|\frac{5\left(\left(\frac{4}{5}\right)^{k+1}+1\right)}{3\left(\left(\frac{4}{5}\right)^k+1\right)}\right| = \frac{5}{3}.$$

(e) We apply the ratio test, thus

$$R=\lim_{k\to\infty}\left|\frac{a_k}{a_{k+1}}\right|=\lim_{k\to\infty}\left|\frac{2^k z^{2k}}{k^2+k}\frac{(k+1)^2+k+1}{2^{k+1}z^{2(k+1)}}\right|=\lim_{k\to\infty}\left(\frac{k+2}{k}\right)\frac{1}{2|z|^2}=$$

$$=\lim_{k\to\infty}\left(\frac{1+2/k}{1}\right)\frac{1}{2|z|^2}=\begin{cases}0, & 2|z|^2<1\\ \infty, & 2|z|^2>1\end{cases}.$$

The convergence radius $R=1/\sqrt{2}$.

(f) We apply the ratio test, thus

$$R=\lim_{k\to\infty}\left|\frac{a_k}{a_{k+1}}\right|=\lim_{k\to\infty}\left|\frac{z^{2k}}{4^k k^k}\frac{4^{k+1}(k+1)^{k+1}}{z^{2(k+1)}}\right|=\lim_{k\to\infty}\left|\frac{4(k+1)}{z^2}\frac{(k+1)^k}{k^k}\right|=$$

$$=\lim_{k\to\infty}\frac{4(k+1)}{|z|^2}\left(\frac{1+1/k}{1}\right)^k=\infty.$$

(g) We apply the ratio test, thus

$$R=\lim_{k\to\infty}\left|\frac{a_k}{a_{k+1}}\right|=\lim_{k\to\infty}\left|\frac{k^k}{1+2^k k^k}\frac{1+2^{k+1}(k+1)^{k+1}}{(k+1)^{k+1}}\right|=\lim_{k\to\infty}\left|\frac{k^k}{(k+1)^k}\frac{1+2^{k+1}(k+1)^{k+1}}{(1+2^k k^k)(k+1)}\right|=$$

$$=\lim_{k\to\infty}\left|\left(\frac{1/k}{1+1/k}\right)^k\left(\frac{1}{(1+2^k k^k)(k+1)}+\frac{2^{k+1}(k+1)^k}{1+2^k k^k}\right)\right|=$$

$$=\lim_{k\to\infty}\left|\left(\frac{1/k}{1+1/k}\right)^k\left(\frac{1}{(1+2^k k^k)(k+1)}+2\left(\frac{k+1}{k}\right)^k\frac{1}{1/(2^k k^k)+1}\right)\right|=2.$$

5. Find the maximum and minimum absolute value of z^2-z in the disc: $|z|\le 1$.

Solution

The maximum of $\left|z^2-z\right|$ must occur on the boundary of the disk ($|z| = 1$).

The latter can be parametrized $z=e^{it}$, $0\le t\le 2\pi$; whence

$$\left|z^2-z\right|^2=\left|z-1\right|^2=(e^{it}-1)(e^{-it}-1)=1-e^{it}-e^{-it}+1=2-2\cos t.$$

Thus the maximum of $\left|z^2 - z\right|$ is 2, which occurs at $t = \pi$.

The minimum of $\left|z^2 - z\right|$ is 0, which occurs at $z = 0$.